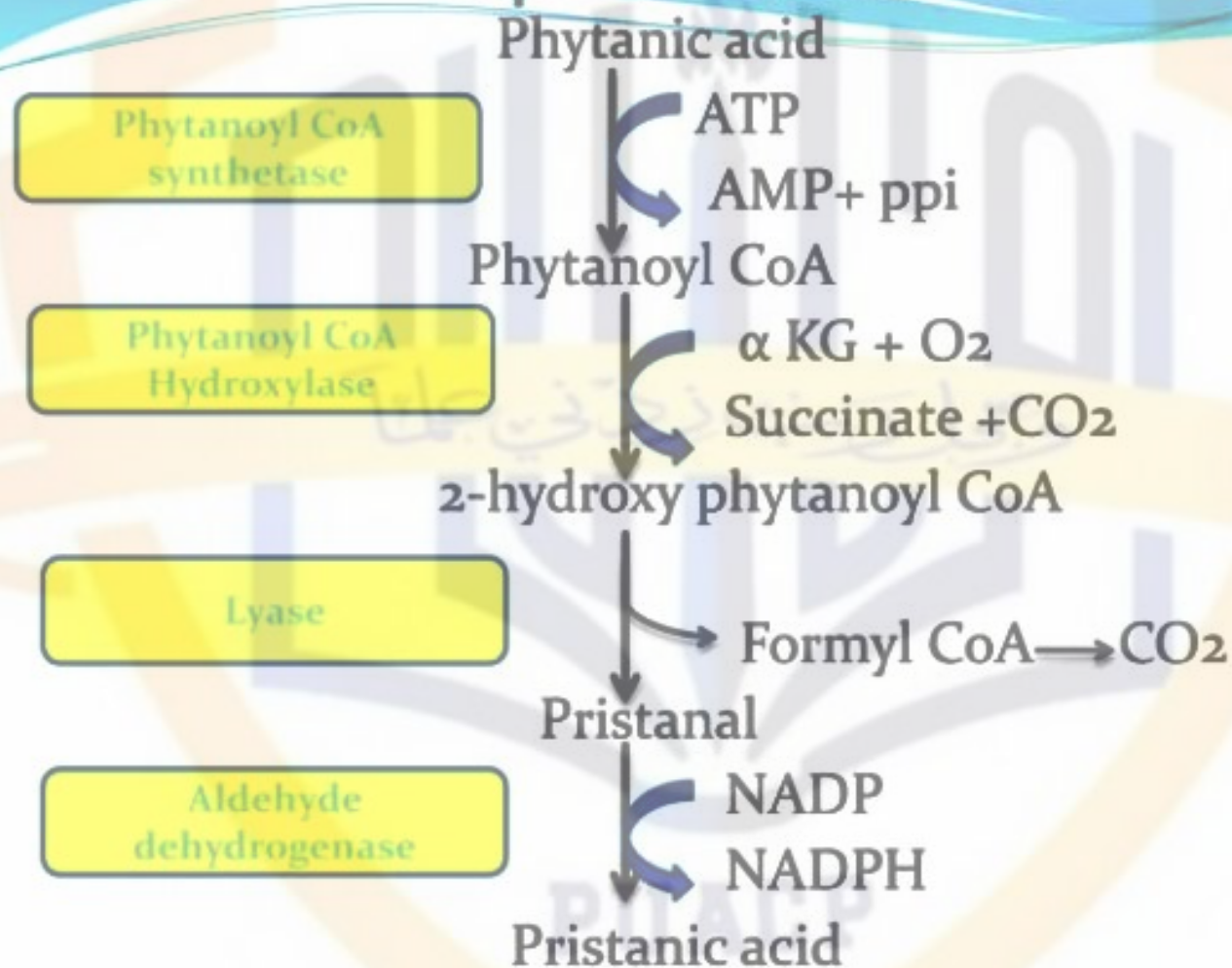


Alpha - oxidation

- *Defined as the oxidation of fatty acid (methyl group at beta carbon) with the removal of one carbon unit adjacent to the α carbon from the carboxylic end in the form of CO_2*
- *Alpha oxidation occurs in those fatty acids that have a **methyl group**(CH_3) at the beta-carbon, which blocks beta oxidation.*
- *Substrate:-**Phytanic acid**, which is present in milk or derived from phytol present in chlorophyll and animal fat*
- ***peroxisomes** is the cellular site.*
- *No production of ATP*

Alpha Oxidation



Beta- Oxidation

- **Beta –oxidation** is defined as the oxidation and **splitting of two carbon units at beta carbon atom.**
- This results in sequential removal of 2 carbon fragments as **acetyl CoA** until the complete oxidation of fatty acids.
- Beta oxidation occurs almost in all tissues except, erythrocytes and adrenal medulla.

Steps of Beta-oxidation proper

